

T.C. ISTANBUL AYDIN UNIVERSITY SOFTWARE ENGINEERING (ENGLISH) GRADUATION PROJECT –2 INTERIM REPORT FORM			
Student's Name – Surname:	Gamze Gezgin Seryal Tuncer Beyza Kondu	Signature	
Student Number:	B2105.090146 B2305.090197 B2205.010086		
Advisor's Name – Surname:	Assoc. Prof. (Ph.D.) Faruk BULUT	Signature	
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<i>Suspicious and Abnormal Human Activity Detection in Surveillance Systems Using AI-Based Models</i>			

1. SUMMARY OF THE WORK CARRIED OUT DURING THE SEMESTER

This interim report presents the experimental work conducted for the graduation project titled "Suspicious and Abnormal Human Activity Detection in Surveillance Systems Using AI-Based Models". The objective is to develop and compare deep learning models capable of automatically detecting three categories of human activity in surveillance video footage: Fighting, Vandalism, and Normal Activities.

The work was structured in four main phases:

- Dataset preparation: Four publicly available datasets (UCF-Crime, DCSASS, Real Life Violence Situations, KTH) were integrated and mapped to three target classes.
- Baseline experiments (Exp. 1.1, 1.2): VGG16+LSTM and MobileNetV2+LSTM were trained on UCF-Crime only to establish baseline performance.
- Extended dataset experiments (Exp. 2, 3.1, 3.2): The same CNN backbone was retrained on the combined four-dataset corpus, and three temporal modules (LSTM, GRU, BiLSTM) were compared under identical conditions.
- YOLOv8 integration (Exp. 4): A person-detection pre-processing step using YOLOv8n was added to the best-performing pipeline to evaluate the effect of region-of-interest (ROI) cropping on accuracy.

2. LITERATURE REVIEW

Automatic violence and anomaly detection in video surveillance has been an active research area for over a decade. Early approaches relied on handcrafted spatiotemporal descriptors such as MoSIFT and dense trajectories [1]. The introduction of deep learning significantly advanced the field; convolutional neural networks (CNNs) replaced manual feature engineering by learning spatial representations directly from raw pixels [2].

To capture temporal dynamics, researchers combined CNNs with recurrent architectures. Patel [3] proposed a hybrid ResNet-50 + LSTM model and demonstrated its effectiveness on the Hockey Fights, Violent Flows, and Movie Fights datasets. Similarly, Sudhakaran and Lanz [2] introduced a two-stream CNN-LSTM architecture that processes appearance and motion streams in parallel, achieving competitive results on multiple benchmarks while maintaining a lower parameter count than 3D convolutional approaches.

The choice of recurrent module has also been studied. Gated Recurrent Units (GRUs) offer a simplified gating mechanism with fewer parameters than LSTMs and have been shown to achieve comparable accuracy on short-to-medium length sequences [4]. Bidirectional LSTMs (BiLSTMs), which process sequences in both temporal directions, have demonstrated superior performance in tasks where context from both past and future frames improves classification, such as distinguishing the onset and resolution of a violent event [5].

MobileNetV2, a lightweight CNN architecture based on depthwise separable convolutions, has proven effective as a transfer learning backbone for surveillance applications. Kommanduri and Ghorai [6] used MobileNetV2 as a feature extractor for video anomaly detection, fine-tuning it on annotated data and achieving strong generalization. In a related study, Jaouedi et al. [4] confirmed that combining MobileNetV2 with recurrent layers yields a strong accuracy-efficiency trade-off for activity recognition on embedded and edge platforms.

Object detection has increasingly been integrated into activity recognition pipelines as a preprocessing step. Jocher et al. [7] introduced YOLOv8, a real-time single-stage detector that achieves state-of-the-art speed-accuracy trade-offs on the COCO benchmark. Subsequent works have applied YOLOv8 to surveillance tasks: Gawande et al. [8] combined YOLOv8 with an LSTM classifier for human anomaly detection in CCTV footage, reporting improved precision when person bounding boxes were used to focus the temporal model on the relevant subject. However, the benefit of ROI cropping has been shown to be class-dependent; for events where the scene context rather than a specific person is the primary cue (e.g., arson or explosion), full-frame representations are more informative [9].

Dataset diversity has also been identified as a critical factor. Models trained on a single dataset often fail to generalize; combining multiple sources such as UCF-Crime [10], DCSASS, Real Life Violence Situations, and KTH substantially improves cross-domain robustness, as confirmed by the experimental results in this study.

3. MATERIALS AND METHODS USED IN THE PROJECT

3.1 Material List

Hardware & Platform:

- Kaggle Notebook with NVIDIA Tesla T4 GPU (16 GB VRAM)
- Python 3.12, TensorFlow 2.x, Keras, OpenCV, NumPy, Scikit-learn, Matplotlib, Seaborn
- Ultralytics YOLOv8 (yolov8n.pt, pretrained on COCO) [7]

Datasets:

- UCF-Crime [10]: Pre-extracted PNG frames. Up to 45 videos per class.
- DCSASS: MP4 clips. Fighting & Assault → Fighting class; Robbery, Arson, Burglary, Explosion → Vandalism class.
- Real Life Violence Situations: MP4 videos. Violence → Fighting; NonViolence → NormalVideos.
- KTH Action Recognition: Boxing → Fighting; Jogging, Walking, Handwaving → NormalVideos. Up to 20 videos per source.

3.2 Methods

Preprocessing Pipeline:

- Each video was sampled into exactly 30 equally spaced frames (SEQ_LEN = 30).
- All frames were resized to 64×64 pixels and normalized to [0, 1].
- Train/test split: 75% / 25%, stratified by video (not frame) to prevent data leakage.
- Data augmentation applied only to the training set (rotation ±20°, zoom 20%, horizontal flip, brightness ±20%, shear 10°, channel shift 20), doubling the training set size.

Model Architectures:

- Exp. 1.1 — VGG16 + LSTM: VGG16 (ImageNet weights, frozen) via TimeDistributed + GlobalAveragePooling2D, LSTM(64), Dropout(0.3), Dense(3, softmax).
- Exp. 1.2 — MobileNetV2 + LSTM: Same structure with MobileNetV2 backbone (3.4M vs 138M parameters).

- Exp. 2 — MobileNetV2 + LSTM on Extended Data: Architecture identical to Exp. 1.2, retrained on the four-dataset corpus.
- Exp. 3.1 — MobileNetV2 + GRU: LSTM replaced with GRU(64) for faster training and smaller model size [4].
- Exp. 3.2 — MobileNetV2 + BiLSTM: LSTM replaced with Bidirectional(LSTM(64)) for richer temporal context [5].
- Exp. 4 — YOLOv8 + MobileNetV2 + BiLSTM: YOLOv8n detects persons per frame; the largest bounding box is cropped and resized to 64×64 before being passed to the BiLSTM classifier [7][8]. Falls back to the full frame when no person is detected.

Training Configuration (all experiments):

- Optimizer: Adam | Loss: Categorical Cross-Entropy | Batch size: 4 | Max epochs: 30
- Early stopping: patience = 10, monitored on val_loss, best weights restored
- Reproducibility: NumPy, Python random, and TensorFlow seeds all set to 3

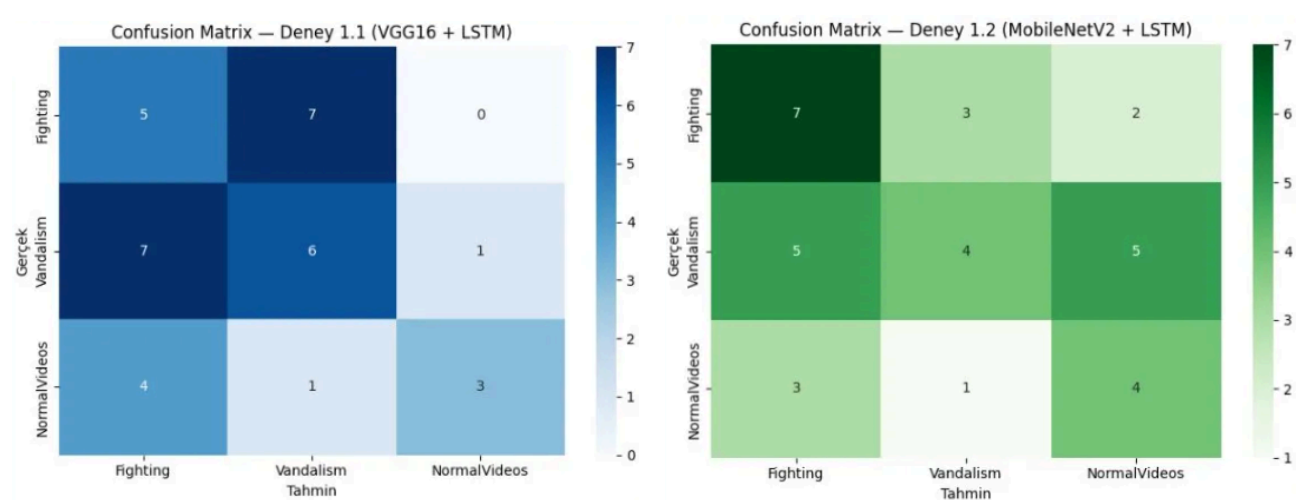
4. RESULTS

Table 1 summarizes the performance metrics for all six experiments.

Experiment	Architecture	Dataset	Acc%	F1	FPS	MB
Exp. 1.1	VGG16 + LSTM	UCF-Crime	41.18	0.4202	5.79	57.9
Exp. 1.2	MobileNetV2 + LSTM	UCF-Crime	44.12	0.4318	5.27	13.1
Exp. 2	MobileNetV2 + LSTM	Extended	95.00	0.9500	5.06	13.1
Exp. 3.1	MobileNetV2 + GRU	Extended	95.00	0.9501	4.97	12.1
Exp. 3.2	MobileNetV2 + BiLSTM	Extended	96.67	0.9667	4.86	17.1
Exp. 4	YOLOv8 + MobileNetV2 + BiLSTM	Extended	72.34	0.7235	4.82	13.1

Table 1. Comparison of all experiments. Best accuracy highlighted in green.

Figures 1–4 provide visual summaries of the experimental results.



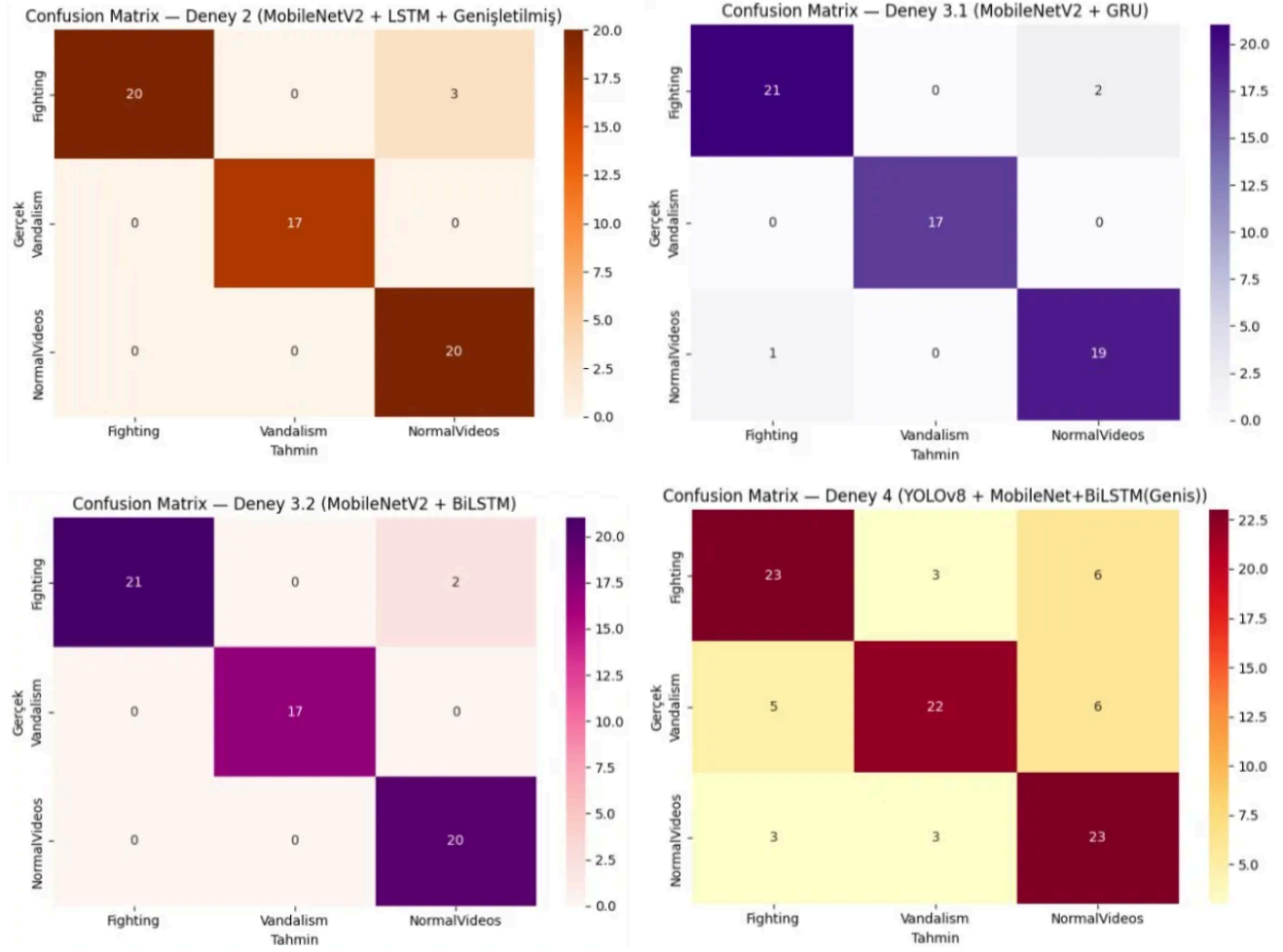
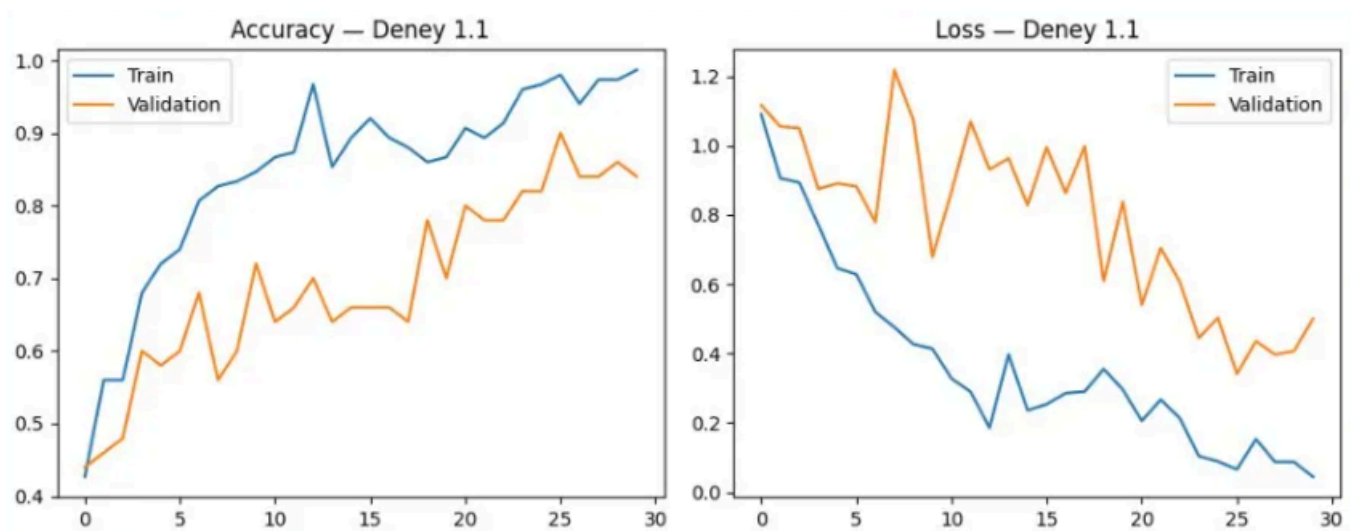
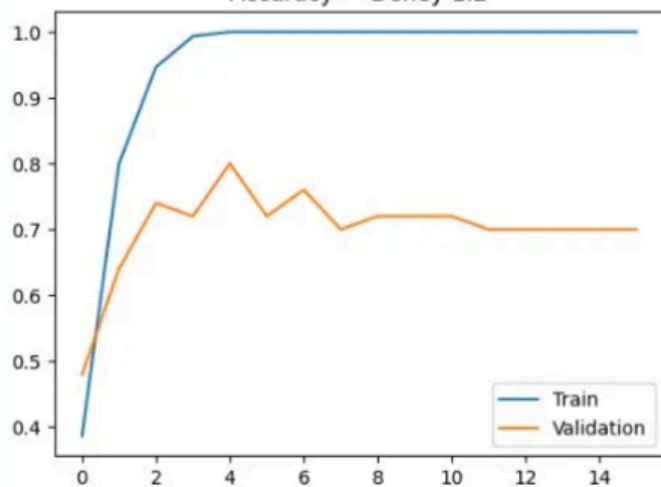


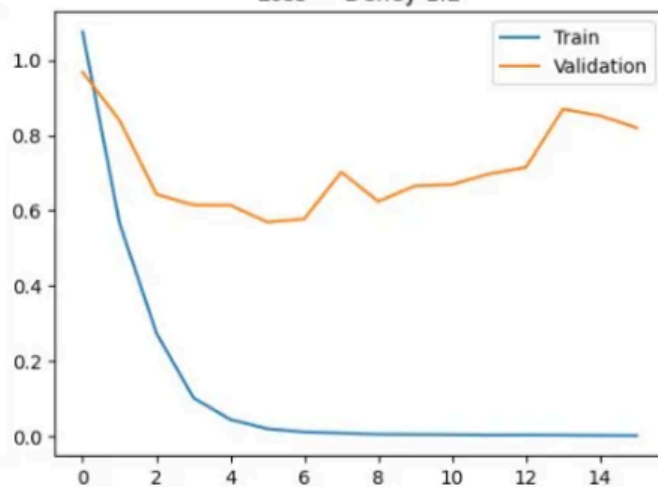
Figure 1. Confusion matrices for all six experiments.



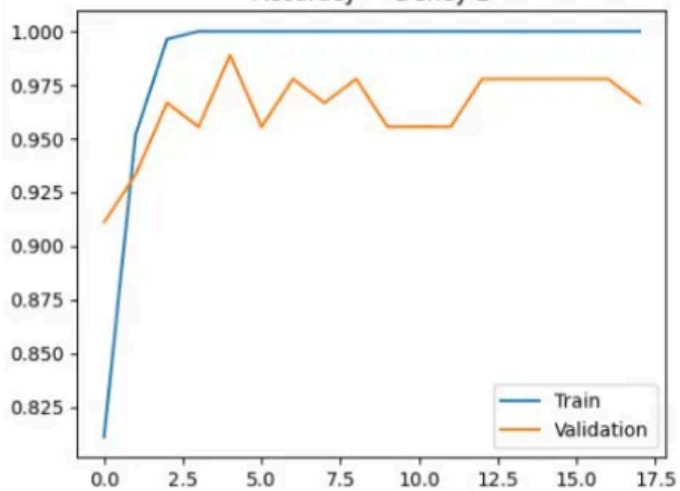
Accuracy — Deney 1.2



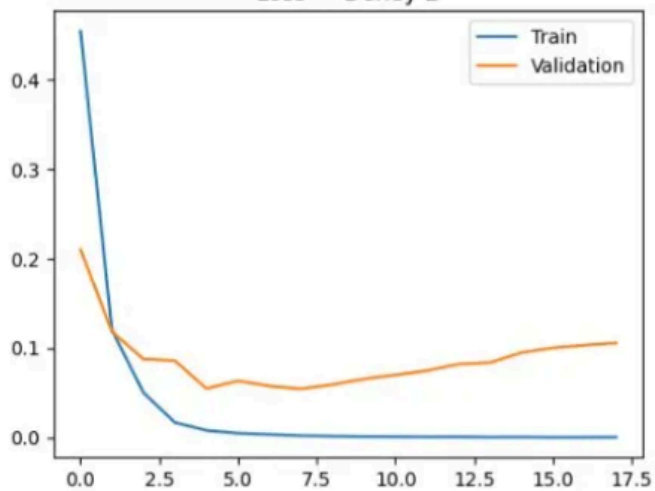
Loss — Deney 1.2



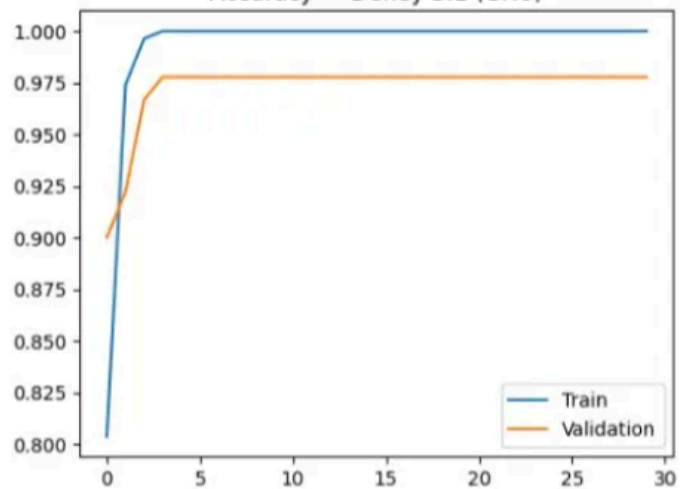
Accuracy — Deney 2



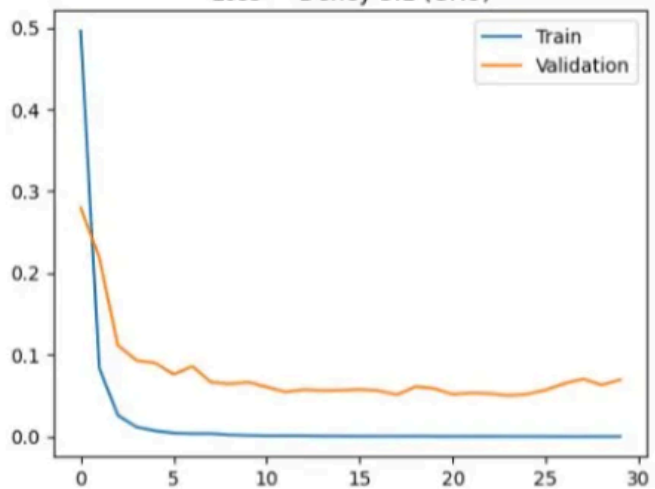
Loss — Deney 2



Accuracy — Deney 3.1 (GRU)



Loss — Deney 3.1 (GRU)



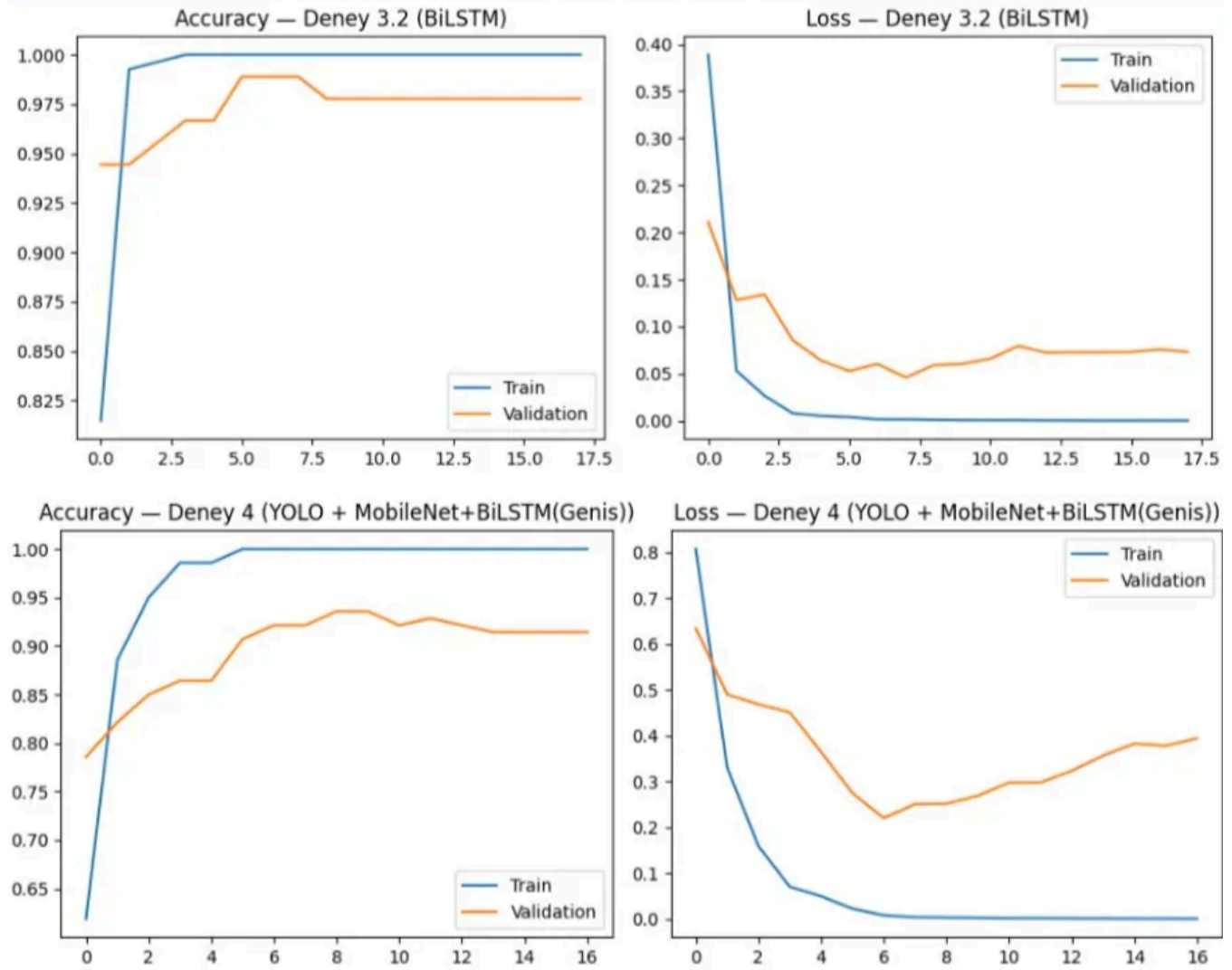


Figure 2. Learning curves (accuracy and loss) for all experiments.

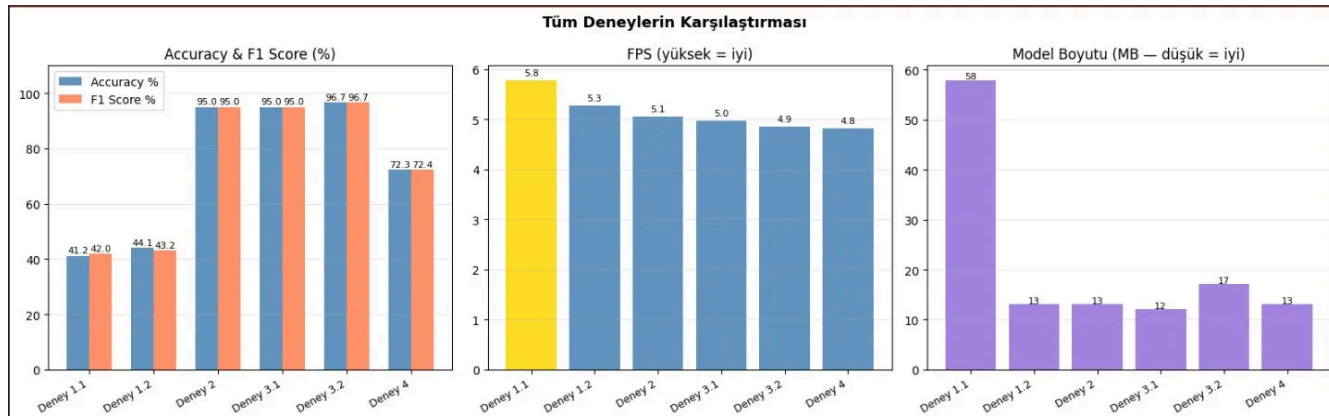


Figure 3. Comparative bar charts for accuracy, FPS, and model size.

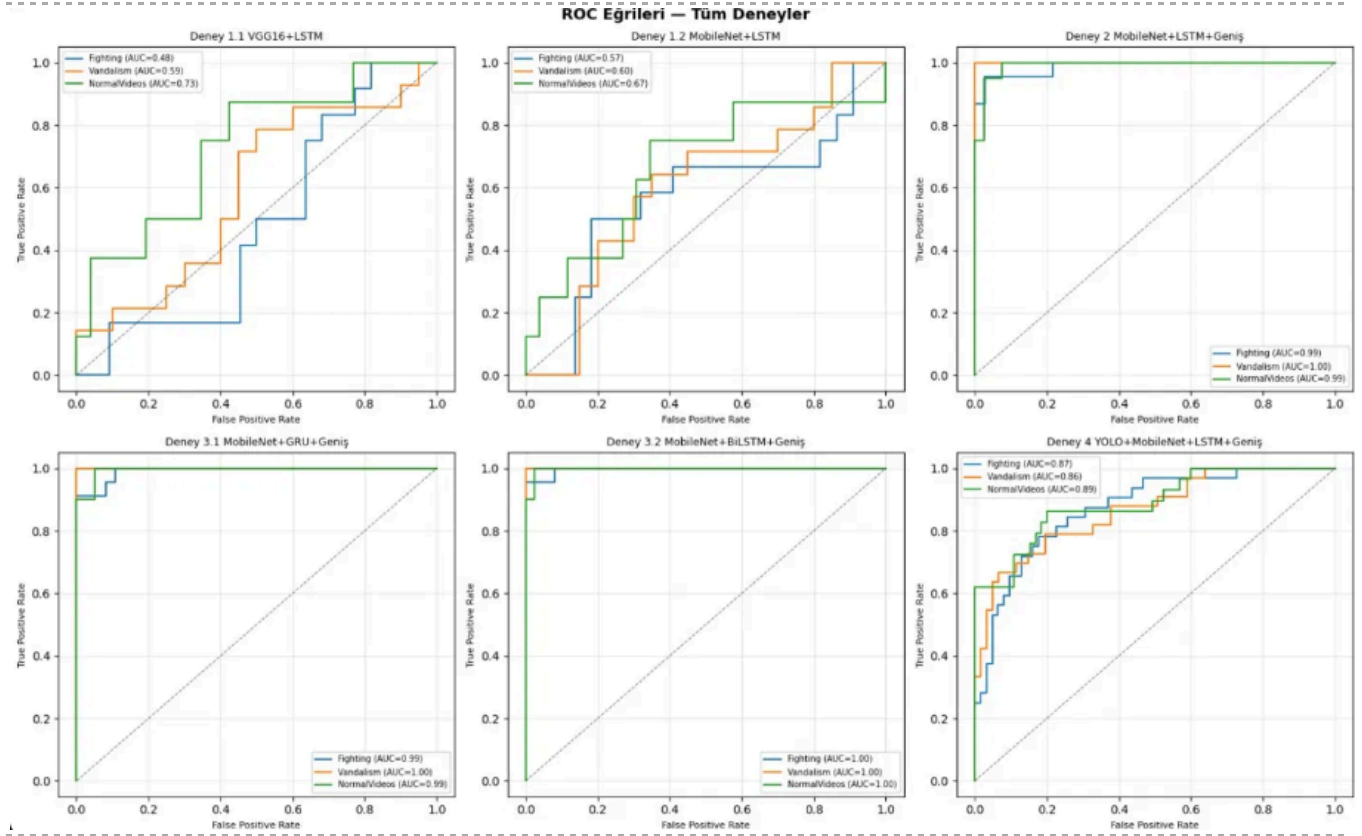


Figure 4. ROC curves per class for all six experiments.

5. DISCUSSION

The results reveal two primary findings. First, dataset diversity has a far greater impact on model performance than architectural choice. The accuracy jump from approximately 42% (UCF-Crime only) to 95% (extended dataset) with the same MobileNetV2+LSTM architecture demonstrates that a richer and more varied training corpus is the most critical factor for generalization in video-based activity recognition. This aligns with findings in the literature that emphasize the importance of multi-source training data for surveillance models [10].

Second, among the temporal modules tested under identical conditions (Exp. 2, 3.1, 3.2), BiLSTM achieved the highest accuracy (96.67%). Processing the 30-frame sequence in both temporal directions allows the model to capture motion patterns from both the beginning and the end of an event, which is beneficial for distinguishing subtle differences between Fighting and NormalVideos. GRU matched LSTM in accuracy (95.00%) while producing a smaller model (12.1 MB vs 13.1 MB), confirming its suitability as a more efficient alternative for deployment-constrained scenarios [4].

The underperformance of the YOLOv8 pipeline (Exp. 4, 72.34%) can be attributed to three factors. First, the Vandalism class includes events such as arson, explosion, and robbery, where the semantically relevant region is the scene itself rather than a person bounding box. Cropping to the detected person discards this contextual information. Second, YOLOv8n occasionally fails to detect persons in crowded or occluded scenes, triggering the full-frame fallback and creating inconsistency within training batches. Third, the mixing of cropped and uncropped frames introduces label noise that complicates optimization. These observations are consistent with prior work showing that ROI-based approaches are most beneficial for person-centric activities and less so for scene-level events [8][9].

Overall, MobileNetV2 + BiLSTM trained on the extended four-source dataset is the recommended configuration, balancing high accuracy (96.67%), acceptable model size (17.1 MB), and practical inference speed (4.86 FPS on a T4 GPU). Future work could explore class-conditional ROI strategies that apply YOLO cropping selectively for the Fighting class while retaining full-frame inputs for Vandalism.

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